INCH-POUND

MIL-C-71150 (AR)
AMENDMENT 4
9 September 1998
SUPERSEDING
AMENDMENT 3
12 November 1997

MILITARY SPECIFICATION

CARTRIDGE, 60MM PRACTICE, SHORT RANGE, M766
LOADING, ASSEMBLING AND PACKING

This amendment forms a part of MIL-C-71150 (AR), dated 1 November 1993, and is approved for use by the U.S. Army Armament Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

DRAWINGS, Delete: "6.5", and substitute: "6.4".

Delete: "QUALITY ASSURANCE PROVISIONS ... QAP C12961001 - Dud Plug".

PAGE 4

* 3.3.7.2 Delete in its entirety.

3.4.4: Delete in its entirety and substitute:

"3.4.4 Range and deflection precision. The cartridge shall have a horizontal range probable error (PEr) less than 15 meters for all zones when calculated to the nearest one tenth (0.1) meter. The deflection probable error (PEd) shall be less than 15 meters for all zones when calculated to the nearest one tenth (0.1) meter)."

AMSC N/A FSC 1315

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3.4.9.1: Delete in its entirety and substitute:

"3.4.9.1 Functioned round. All dud plug heads (regardless of stems) shall blow out of the projectile body upon fuze function."

3.4.9.2: Delete in its entirety and substitute:

"3.4.9.2 Inert round. No dud plug head (regardless of stems) shall exit the projectile body during ballistic flight or upon impact."

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4.3.3, delete requirement "b." in its entirety and substitute: "b. Three or more duds occur."

Delete "g." and "h." in their entirety and substitute:

"g. If the weapon is damaged by abnormal wear of the interior or of the firing pin, or caused by excessive pressure.

h. Two or more improper impact signatures occur."

Add the following:

"i. All three dud plug heads (regardless of stems) are missing from the projectile body in the inert fuze ballistic test.

j. More than one dud plug head (regardless or stems) remains in a single projectile body upon fuze function except in the case of a particular round that was an actual dud. In this case, paragraph "i." Above applies."

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* Table I, Delete under Dud Plug and subsitute the following:

"Dud Plug ...(dwg. 12988971)

Examination for defects ...10(a) ... 3.1 ... 4.4.2.9

Add Note: "(a) Ten (10) samples are required from each mold cavity."

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4.4.2.1, Major 105: Delete:

"Length of large outside diameter section",

and substitute:

"Length between large outside diameter aft face to firing pin point".

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4.4.2.3, Major 102: Delete the word "excessive" and substitute the word "inadequate".

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4.4.2.5: Delete Specials A, B, C, D, E, F and G in their entirety and add the following:

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
"1	Fuze striker in armed position	100%	3.3.4	Gage
2	Breech plug assembly firing pin not below flush after assembly to round	100%	3.3.3	Gage
3	Powder from energetic cartridges found on outside of round	100%	3.1	Visual
Major				
108	Obturator ring missing	100%	3.3.2	Visual
109	Fin tail cracked, damaged or distorted	100%	3.1	Visual
110	Fuze packing clip missing or improperly assembled	100%	3.1	Visual
111	Cartridge assembly fails to fully enter complete round alignment gage	100%	3.3.1	Gage (b)
112	Any dud plugs missing	100%	3.3.7	Visual"

Under Notes: Delete "(a)" and substitute "(b)".

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4.4.2.6: Delete Special A in its entirety.

Add the following:

“Major 101 Packing clip missing from fuze ... 100% ... 3.2 ... Visual”

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* 4.4.2.9 Delete in its entirety and substitute:

PARAGRAPH	TITLE	SHEET 1 of 1		DRAWING NUMBER 12988971
"4.4.2.9	Dud Plug			NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR				
101	Outside diameter of plug head	(a)	3.1	Gage
102	Thickness of plug head	(a)	3.1	Gage
103	Outside diameter of plug stem	(a)	3.1	Gage
104	Inside diameter of plug stem	(a)	3.1	Gage
105	Length from head flat to end of stem	(a)	3.1	Gage
MINOR				
201	Poor workmanship	30	3.6	Visual
202	Stem slit missing	(a)	3.1	Visual

NOTES: (a) 20 samples from each mold cavity shall be examined for this feature or dimension. The sample of 20 shall incude the first two parts from the beginning of each production run and the last two parts from the end of the run. The remaining samples of (at least sixteen) shall be randomly selected during the production run. Any defects found in any of the 20 samples from each die cavity shall be cause for rejection of all dud plugs produced from that particular die cavity."

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4.4.3.1.1: Delete “c.” in its entirety and substitute:

"c. Short rounds, Major defect (see 3.4.2).".

Delete “c.” in its entirety and substitute:

"e. Metal parts separation, Major defect (see 3.4.3).".

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Delete “i.” in its entirety and substitute:

"i. All dud plug heads out of any one projectile in the inert fuze ballistic test, Critical defect (see 3.4.9.2).".

Add the following:

"j. More than one dud plug head (regardless of stems) remains in a single projectile body after fuze function in the live fuze ballistic test, Major defect (see 3.4.9.1)."

4.4.3.1.2: delete requirement "b." in its entirety and substitute: "b. Three or more duds occur."

Delete "i." And "j." in their entirety and substitute:

"i. All three dud plug heads (regardless of stems) are missing from the projectile body in the inert fuze ballistic test.

j. More than one dud plug head (regardless of stems) remains in a single projectile body upon fuze function except in the case of a particular round that was an actual dud. In this case, paragraph "i." above applies.

4.5.1.1, first paragraph, after second sentence: Add the following: “All rounds shall be fired onto a flat, nongrassy earthen surface that is not marshy or frozen.”

Add the following:

"4.5.1.3: Accuracy. The individual defelctions and ranges shall be recorded to an accurancy of both to 0.01 meter. Range probable error shall be calculated to the nearest 0.1 meter."

* 4.5.5, Delete in its entirety.

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5.3.1: Delete: “12944136”, and substitute: “12953578”.

6.9 Add the following:

“6.9 Inspection lot definition. A homogeneous collection of units of product or one shipment of component parts or assemblies from a subcontractor, bearing identification and treated as a unique entity from which a sample is to be drawn and inspected (and tested) to determine conformance with the acceptability criteria.”

6.10 Delete in its entirety.

6.11 Add the following:

"6.11 Outlier test. When the accept criteria is not met and if it is suspected that a statistical outlier may be present in the sample of 30 observations. Arrange the observed values in order of increasing magnitude (X1, X2 .. Xn-1, Xn. Using the statistics calculated for the sample, calculate Tn where:

$$Tn = \frac{Xn - \bar{X}}{s}$$

If X1, the smallest value, rather than Xn, is the doubtful value the test criterion is:

$$T1 = \frac{\bar{X} - X1}{s}$$

Where:

$\bar{X}$ = arithmetic average of all n values
s = estimate of the population standard deviation based on the sample data calculated as follows:

$$s = \left[\frac{\sum\left(x_i - \bar{x}\right)^{1/2}}{n-1}\right] = \left[\frac{n\sum x_i^2 - \left(\sum x_i\right)^2}{n(n-1)}\right]^{1/2}$$

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If Tn is less than or equal to 2.745, the highest value may not be eliminated. The lot is rejected. If Tn is greater than or equal to 2.745, the largest value obtained in the sample shall be eliminated and the sample statistics X and s shall be recalculated using the twenty-nine remaining values. This shall be compared to the accept criteria. If the criteria is met, accept the lot. If not, the same procedure shall be followed for the Xn-1 observation. Calculate Tn-1 based on the new statistics of the remaining sample of 29. If Tn-1 is less than or equal to 2.730, the lot is rejected. If Tn-1 is greater than or equal to 2.730, then this observation shall be eliminated and the sample statistics calculated based on the remaining 28 observations. If these values meet the accept criteria, accept the lot. If the accept criteria is still not met, reject the lot."

6.12 Add the following:

"6.12 Marshy ground. Marshy ground is defined as an earthen surface that contains standing water or free flowing (fluid) mud or a combination of both."

6.13 Add the following:

"6.13 Frozen ground. Frozen ground is defined as earthen ground within which water has frozen creating a hard surface."

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Army-AR

(Project 1315-0216)

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